



Armada Kalamunda Group

Venous Thromboembolism Prevention Guideline – Adult

1. Purpose

The purpose of this guideline is to ensure to support clinicians implement the delivery of high-quality care to prevent venous thromboembolism (VTE) acquired in hospital and following hospital discharge in accordance with the quality statements outlined in the [Venous Thromboembolism Prevention Clinical Care Standard](#) (ACSQHC, 2018).

2. Background

VTE involves the formation of a blood clot within the deep veins, most commonly of the legs or pelvis (deep venous thrombosis or DVT). These blood clots may become dislodged and then obstruct the pulmonary artery or one of its branches (pulmonary embolism or PE). VTE is a significant preventable adverse event for hospitalised patients.

The incidence of developing a VTE has been shown to be 100 times greater among hospitalised patients than those in community². Serious adverse outcomes resulting from VTE may occur, including an increased risk of recurrent thrombosis, morbidity from post-thrombotic syndrome or death.

3. Applicability

This guideline applies to medical officers, nurses, midwives and pharmacists. This guideline does not include clinical practice advice for paediatric and neonatal patients.

Nursing and midwifery staff and pharmacists are to be aware of VTE risk and assist in ensuring the processes for prevention are implemented.

4. Identifying patients for assessment

4.1 Emergency Department

Adult patients to be discharged home from the Emergency Department who, as a result of their acute illness or injury, have significantly reduced mobility relative to normal state must undergo a VTE risk assessment and be prescribed appropriate prophylaxis by an Emergency Department clinician prior to leaving the Emergency Department.

All other patients to be discharged home from an Emergency Department do not need to be assessed for VTE risk.

4.2 Admitted patients

All adult patients admitted to Armada Kalamunda Group (AKG) must undergo a VTE risk assessment within 24 hours of admission and, if appropriate, be prescribed pharmacological and/or mechanical prophylaxis.

4.3 Pregnant and post-partum women

All pregnant and post-partum women must undergo a VTE risk assessment:

- During the first antenatal review and/or during booking
- During admission into a non-obstetric setting for a non-pregnancy related complaint
- During admission into an obstetric setting for a pregnancy or non-pregnancy related complaint
- Immediately after birth (postnatal period).

Please see the [Venous Thromboprophylaxis in Pregnancy and Post-Partum Guideline \(OBS-GUI-0462-14\)](#) for obstetric-specific guidance.

4.4 Planned admission

Patients undergoing planned surgical and invasive interventions and/or imaging guided invasive interventions:

- Prior to planned surgery and invasive procedures, and/or imaging guided invasive procedures, medical officers must assess the risks and benefits of stopping pre-existing, established anticoagulation or anti-platelet therapy before discontinuing these therapies.
- Prophylaxis should be considered for day surgery patients based on evidence in situations of significantly reduced mobility relative to normal state, prolonged anaesthesia and for patients demonstrating one or more other risk factors.³

5. Risk assessment

5.1 Assessing VTE risk

Medical Officers should use the risk assessment criteria outlined in [Appendix 1](#) and ensure that the following steps are undertaken during the assessment:

1. Assess baseline (intrinsic) risk
2. Assess additional risks posed by hospitalisation or illness (extrinsic risk)
3. Assess contraindications to:
 - a. Pharmacological prophylaxis
 - b. Mechanical prophylaxis
4. Formulate overall risk assessment outcome (consider risk of prophylaxis against benefit) and document on the Hospital Medication Chart in the VTE Risk Assessment section
5. Decide if VTE prophylaxis is required
6. Select the form of prophylaxis to be used based on the risk assessment
7. Inform patient and/or carer of the VTE prophylaxis measures to be undertaken – see Section 9: [Engaging the patient and/or carer](#) for further information
8. Reassess prophylaxis regularly:
 - a. Minimum every seven (7) days
 - b. If the patient's condition changes; and
 - c. At transfer of care.

5.2 Assessing VTE risk in pregnant and post-partum women

- The VTE risk assessment for a pregnant woman admitted into a non-obstetric setting for a non-pregnancy related complaint can be documented on the Hospital Medication Chart.
- Any other tools used to assess VTE risk in a non-obstetric setting must identify all pregnant and post-partum women to be at risk of VTE.
- The decision to commence pharmacological and/or mechanical prophylaxis must be referred to a Consultant Obstetrician.
- The following tools are approved for use across the AKG to assess pregnant and post-partum women in an obstetrics setting:
 - Antenatal Venous Thromboembolism (VTE) Risk Assessment Tool (AKMR23.1)
 - Postnatal Venous Thromboembolism (VTE) Risk Assessment Tool (AKMR23.2).

For further clinical practice guidelines associated with VTE prevention for obstetrics patients please refer to the [Venous Thromboprophylaxis in Pregnancy and Post-Partum Guideline \(OBS-GUI-0462-14\)](#).

5.3 Documenting VTE risk

Medical Officers must document in the patient's medical record the outcome of the risk assessment on an approved clinical record such as:

- WA Hospital Medication Chart (short and long stay versions), which includes a dedicated VTE risk assessment section
- Antenatal Venous Thromboembolism (VTE) Risk Assessment Tool (AKMR23.1)
- Postnatal Venous Thromboembolism (VTE) Risk Assessment Tool (AKMR23.2)
- Integrated Progress Notes.

5.4 Minimum VTE prevention strategies

Irrespective of a patient's VTE risk outcome, the following measures should be observed during the episode of care:

- Patients must remain adequately hydrated (unless contraindicated due to their clinical condition e.g. fluid restriction due to congestive cardiac failure) and must be encouraged to mobilise as soon as possible and to continue being mobile post discharge.⁴
- A plan for early mobilisation should be developed by a multidisciplinary team with the patient and their family/carer.

6. Prescribing appropriate prophylaxis

[Appendix 2](#) is a clinical decision support tool for Medical Officers when prescribing prophylaxis in accordance with the Therapeutic Guidelines.

6.1 Pharmacological prophylaxis

It is recommended to continue pharmacological prophylaxis until the patient regains normal mobility or is transferred to home or other care setting. In some instances pharmacological prophylaxis may need to be continued beyond the hospital stay.

The decision to commence pharmacological prophylaxis should be made after considering the benefits of treatment i.e. reducing VTE risk against the risk associated with treatment (bleeding and other contraindications).

6.1.1 Individualising the dose of pharmacological prophylaxis

Consideration:

Some agents are contraindicated or require a dose reduction in the elderly patient or those with renal impairment.

- Prescribers should refer to current product information to select a safe dose for individual patients, taking care to select the dose recommended for prophylaxis and not the dose recommended for therapeutic anticoagulation.
- Obese patients (body mass index $>30\text{kg/m}^2$) may have an increased risk of VTE⁵ and may require higher pharmacological prophylaxis doses. Evidence is limited regarding the recommended dosing regimen; therefore clinician discretion is warranted, taking into account both the risk of clotting and the risk of bleeding.
- Patients with renal impairment often require a reduced dose of pharmacological prophylaxis; or depending on level of impairment, one agent may be preferred over the other. Renal impairment should be calculated using creatinine clearance as the unit of measure before decisions regarding pharmacological prophylaxis are made.
- Decisions about appropriate concomitant use of these medications for VTE prophylaxis should be made on an individual patient basis in consultation with the Consultant Medical Officer.

6.1.2 Drug interactions

Drugs like anticoagulants (e.g. warfarin, dabigatran), anti-platelet agents, selective and non-selective non-steroidal anti-inflammatory drugs, and antithrombotic agents may interact with prophylactic agents to increase the risk of bleeding³. Other drugs may interact with prophylactic agents to increase bleeding risk or reduce their efficacy. Check carefully and refer to product information and/or appropriate clinical references.

Decisions about appropriate concomitant use of these medications for VTE prophylaxis should be made on an individual patient basis in consultation with the Consultant Medical Officer.

6.1.3 Contraindications to pharmacological prophylaxis

Contraindications to pharmacological prophylaxis must be considered before a patient is prescribed therapy.

Contraindications can be either absolute or relative. Where an absolute contraindication exists, the use of pharmacological prophylaxis should be avoided due to life-threatening risk, while relative contraindications require caution to be exercised and the benefits of therapy to be weighed against the risk.

Absolute contraindications may include (but are not limited to):

- Known hypersensitivity to agents used in pharmacological prophylaxis
- History of or current, heparin-induced thrombocytopenia (alternative agents available)
- Active bleeding or risk of clinically significant bleeding.

Relative contraindications may include (but are not limited to):

- Any risk of bleeding

- Creatinine clearance <30mL/minute (reduced dosages or specific agents may be used in some cases)
- Use of low dose aspirin for prevention or treatment of cardiovascular disease or other antiplatelet therapy.

Where pharmacological prophylaxis is contraindicated, mechanical prophylaxis remains an option and should be considered, as indicated, until the patient is mobile.

6.1.4 Anaesthesia and VTE

The type of anaesthesia a patient receives has been identified as impacting on risk of VTE. Patients receiving regional anaesthesia, have significantly lower rates of DVT compared with those receiving general anaesthesia.

As for all surgery and imaging guided invasive procedures (but particularly if central neural blockade is used), timing of pharmacological prophylaxis should be carefully planned with the anaesthetist to minimise the risk of developing an epidural haematoma.³

The NHMRC² recommends:

- No pharmacological prophylaxis with LMWH should be administered prior to establishment of neural blockade, or the block should be performed ≥12 hours after the last dose of LMWH if preoperative prophylaxis has been administered with this drug. Dosing after surgery should start ≥ 6 hours postoperatively.
- If an epidural catheter has been placed it should be removed:
 - ≥4 hours before a postoperative dose of pharmacological prophylaxis with LMWH; and
 - ≥12 hours after a previously administered LMWH dose.

6.2 Mechanical prophylaxis

At a minimum, all higher risk patients should have mechanical prophylaxis with appropriate pharmacological prophylaxis usually provided until the patient is fully mobile or as indicated.

Further specific guidance is provided in Appendix 3 – [Mechanical VTE prophylaxis for surgical patients in the Same Day Unit](#).

6.2.1 Graduated compression stockings

- All stockings must be fitted by nursing staff and worn correctly according to the manufacturer's recommendations.
- It should be noted that graduated compression stockings may increase the risk of falls in mobilising patients. Approved non-slip over-socks or appropriate footwear should be worn to avoid falls.
- Stockings must be removed daily to assess skin condition and perfusion and to provide skin care – this should be documented on the patient care plan.
- Pressure areas (such as the heel, toes and behind the knee) should be checked twice daily, most preferably at every shift – the outcome of these assessments should be documented in the patient's medical record.

6.2.2 Intermittent pneumatic compression devices

- Intermittent Pneumatic Compression (IPC) devices periodically compress the foot, calf and/or thigh muscles of the leg (inflation pressures 35-40mmHg at approximately 10s/min) which

promotes blood flow.⁷ Foot and calf compression devices may further decrease the risk of VTE when used with other preventative therapies.^{9,10}

- Refer to manufacturer's instructions for correct application of selected device. In particular measurement's to ensure correct garment sizing.
- Remove garments immediately if the patient experiences tingling, numbness or pain. Take down garments at least every eight (8) hours and inspect skin.⁴
- Continuous therapy is most effective. Minimise interruptions to IPC to less than 45 minutes.⁴
- Continue IPC for no less than 72 hours post operatively or until patient is fully ambulant.⁷
- For surgical patients, apply IPC therapy pre-operatively before the patient becomes immobile. Once in theatre the IPC pumps can be attached and continued through the perioperative journey.

6.2.3 Contraindications to mechanical prophylaxis

Compression stockings may be contraindicated in patients with:

- Morbid obesity where correct fitting cannot be achieved
- Inflammatory conditions of the lower leg
- Severe peripheral arterial disease
- Diabetic neuropathy
- Severe oedema of the legs
- Unusual leg deformity
- Allergy to stocking material
- Cardiac failure³
- Lower leg trauma (affected limb).

IPC can exacerbate lower limb ischemic disease and are contraindicated in patients with peripheral arterial disease or arterial ulcers.³

IPC is contraindicated in acute lower limb DVT. The NHMRC notes that a recent study provides no evidence to support the routine use of graduated compression stockings in immobile, hospitalised patients following acute stroke.³

6.2.4 Complications of mechanical prophylaxis

Bunching of the stockings from incorrect fitting could result in leg ulceration, pressure injuries, slipping and falling on mobilisation.²

7. Documentation of prophylaxis

Once the decision is made to prescribe pharmacological and/or mechanical prophylaxis, an order must be written on the Anticoagulation Medication Chart which contains a dedicated VTE section.

Where the Anticoagulation Medication Chart is used, the Medical Officer must prescribe pharmacological and/or mechanical prophylaxis within the dedicated section. Prescribing outside of this section may lead to duplication of orders and risk of patient harm.

Checks associated with mechanical prophylaxis must also be documented at least twice daily by nursing staff/midwives. Checks should be documented on the Patient Care Plan.

8. Other patient groups

8.1 Stroke patients

- For ischaemic stroke, consider low-dose UFH or LMWH in patients with restricted mobility.
- For haemorrhagic stroke, consider low-dose UFH or LMWH 2-4 days after stroke in patients with restricted mobility.
- If anticoagulation is contraindicated patients should be considered for IPC.

8.2 Peri-operative and peri-procedural management of patients receiving regular anticoagulation

For surgery and invasive procedures an important consideration is the timing of stopping and restarting regular anticoagulants (e.g. warfarin), and the timing of prescription of pharmacological prophylaxis to bridge anticoagulation therapy for peri-operative patients. As an example after warfarin therapy is discontinued, it takes several days for the antithrombotic effect to recede. When warfarin is recommenced, several days are required to reach therapeutic anticoagulation levels.

There is no current consensus on the appropriate peri-operative management of anticoagulation. Medical Officers should review current evidence prior to planning treatment.

Medical Officers should review current evidence regarding continuation of regular anticoagulants or antiplatelet therapy where only minor procedures are planned and document any clinical decision-making and management plan in the patient's medical record.

8.3 Patients on medication containing oestrogen

Medical Officers should review current evidence (including risks of unplanned pregnancy vs benefit of VTE prevention) to determine whether patients, prior to, or during admission, should discontinue oestrogen-containing oral contraceptives or hormone replacement therapy if clinically appropriate. In the case of oestrogen-containing oral contraceptives, these risks should be communicated to the patient and if it is thought appropriate to stop oral contraceptives, adequate alternative contraception should be arranged until oral contraceptives are restarted.⁶

8.4 Patients on medication increasing the risk of VTE

There are a number of other medications that may increase the risk of developing a VTE, particularly during periods of immobilisation or after surgery. These may include, but are not limited to, agents such as tamoxifen, epoetin alfa, strontium ranelate, and raloxifene. Medical Officers should consider temporary cessation of such agents or VTE prophylaxis where appropriate.

9. Engaging the patient and/or carer

Patients, carers and their families must be informed about:

- What a VTE is
- Signs and symptoms of VTE
- Risk factors specific to the patient's condition
- Effective interventions to reduce the risk of VTE developing
- Any pharmacological and/or mechanical prophylaxis they are receiving
- VTE prevention discharge plans (where required).

When a treatment decision has been made, clinicians must document that the patient has received an explanation of risks and benefits of prophylaxis. This should be recorded within the patients' healthcare record.

Clinicians must also document when they provide patients and/or carers with additional information regarding VTE prevention.

10. Reassessing VTE risk

After the initial risk assessment, the patient's risk of bleeding and of VTE should be reassessed:

- Regularly as clinically appropriate, at a minimum every seven (7) days
- When clinical condition changes (e.g. unplanned surgery, changes in mobility)
- At transfer of care.³

Reassessment is required to:

- Ensure that appropriate methods of VTE prophylaxis are used
- Ensure that VTE prophylaxis is being used correctly
- Identify adverse events resulting from VTE prophylaxis or its absence.

10.1 Reassessing the risk at discharge and continuity of care

- Patients who have been identified as being at risk of VTE during their admission should be reassessed at the point of discharge. Consideration should be made regarding the need for extended prophylaxis.
- Medical Officers must ensure the development of a prospective action plan for patients requiring continuation of pharmacological and/or mechanical prophylaxis on transfer home or to another care level. The plan is to be communicated in a timely manner to the patient's care provider and explained to the patient/carer/family. This is particularly important when patients are transferred into community or residential aged care.
- The recommended duration of pharmacological prophylaxis will vary depending on the patient's medical status. When determining the duration of prophylaxis, consideration should be given to the patient's mobility status and the clinical evidence related to their specific condition – see Appendix 1 for recommended duration.
- On transfer to home or another care services, a patient's supplies of prophylactic medication should be arranged to enable uninterrupted treatment. Referral to another care model should be arranged including assurance of follow-up and continuity of supply as needed. Patients should understand the reason for ongoing treatment and the anticipated timeframe for discontinuation of the treatment. Patients must receive education on the administration of treatment as needed and be encouraged to mobilise (unless instructions for mobility restriction are in place).

11. Roles and responsibilities

Role	Responsibility
Director Clinical Services	• Ensure education resources are available regarding VTE prophylaxis. • Ensure the availability of VTE risk assessment and prophylaxis decision support tools to all clinical units. • Ensure formulary management includes availability of medications recommended for VTE prophylaxis. • Ensure case review of patients developing a VTE that occurs during, or as a result of, a hospital admission. • Support clinical units to regularly review their VTE data and develop strategies towards improving prophylaxis where required.
Consultant Medical Officer (or delegate)	• Ensure VTE risk assessment is performed on all adult admitted patients within 24 hours, and those discharged from an Emergency Department with significantly reduced mobility relative to normal state. • Review the patient's related bleeding risk and based on that assessment; ensure prescription and administration of appropriate prophylaxis as required. • Discuss the reason for treatment, risks and consequences of VTE prophylaxis with the patient on admission and on transfer to community or home care where required. • Document VTE risk assessment on the Hospital Medication Chart or other locally approved form. • Document VTE prophylaxis prescription on the Anticoagulation Medication Chart. This, and other significant information, including any relevant dosage adjustment can be documented in the patient's healthcare record or other locally approved forms. • Confirm appropriate peri-operative prescription of both pharmacological and mechanical prophylaxis where indicated. • Ensure regular review of VTE risk is performed during the patient care episode, particularly as clinical condition changes, and that prophylaxis is monitored and adjusted accordingly. • Ensure clinical speciality protocols include VTE prophylaxis where appropriate.

12. Legislative context

The following documents are required to give effect to this policy:

- [Performance Management Policy 2019-20 MP0111/19](#)
- [High Risk Medication Policy OD0561/14](#)

13. Supporting information

The following documents support the implementation of this policy:

- [Venous Thromboprophylaxis in Pregnancy and Post-Partum Guideline \(OBS-GUI-0462-14\)](#)
- [Medication Management Procedure \(AKG-PRO-0255-18\)](#)

14. Compliance, monitoring and evaluation

The Clinical Outcomes Committee will be accountable for monitoring compliance with this guideline.

Compliance will be monitored via:

- Patient medical record audit against the National Quality Use of Medicines for Australian Hospitals Set 1: Antithrombotic therapy. This audit is to be conducted at a minimum on an annual basis and more frequently as required.
- Monitoring of trended hospital-acquired complication data related to VTE as presented in the Health Service Performance Report (HSPR) Indicator ref: P2-26 (d) venous thromboembolism.

15. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 1 – Action 1.27: The health service organisation has processes that:
 - a. Provide clinicians with ready access to best-practice guidelines, integrated care pathways, clinical pathways and decision support tools relevant to their clinical practice
 - b. Support clinicians to use the best available evidence, including relevant clinical care standards developed by the Australian Commission on Safety and Quality in Health Care.
- Standard 4 – Action 4.3: Clinicians use organisational processes from the Partnering with Consumers Standard in medication management to:
 - a. Actively involve patients in their own care
 - b. Meet the patient's information needs
 - c. Share decision-making
- Standard 4 – Action 4.13: The health service organisation ensures that information and decision support tools for medicines are available to clinicians.

16. References

1. Australian Commission on Safety and Quality in Health Care. Venous Thromboembolism Clinical Care Standard. Sydney: ACSQHC; 2018
2. Heit JA, Joseph Melton III L, Lohse CM, Petterson TM, Silverstein MD, Mohr DN, et al. Incidence of Venous Thromboembolism in Hospitalized Patients vs Community Residents. Mayo Clin Proc 2001;76:1102-10.
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4. National Institute for Health and Clinical Excellence (NICE) Guidelines. Venous thromboembolism: reducing the risk: Reducing the risk of venous thromboembolism (deep vein thrombosis and pulmonary embolism) in patients admitted to hospital. January 2010.
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8. Sharma L. Venous Thromboembolism: Prevention and Prophylaxis. Joanna Briggs Institute. 01/03/2015.
9. Falck-Ytter Y, Francis C, Johanson N, Curley C, Dahl O, Schulman S, et al. Prevention of VTE in orthopaedic surgery patients. Antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians evidence-based clinical practice guidelines. CHEST. 2012;141(2Suppl):e278S-e325S

17. Glossary

The following definitions are relevant to this policy.

Term	Definition
Clinically significant bleeding	Major haemorrhage that is life threatening and is likely to result in the need for a massive transfusion; haemorrhage of a smaller volume in a critical area or organ which results in patient morbidity or mortality; or chronic blood loss results in patient morbidity or mortality over time.
Deep Vein Thrombosis (DVT)	A blood clot that occurs in the “deep veins” in the legs, thighs or pelvis. • Asymptomatic deep vein thrombosis is defined as painless DVT detected only by screening with fibrinogen scanning, ultrasound, or ascending venography and is often confined to the distal veins. • Symptomatic deep vein thrombosis results from occlusion of a major leg vein and results in leg pain or swelling. It requires specific investigation and treatment which in hospitalised patients may delay discharge, or require readmission to hospital.
Significantly reduced mobility relative to normal state	Refers to patients who are bedbound, or likely to spend a substantial proportion of the day in bed or in a chair due to the clinical condition for which they are being treated, or unable to walk unaided due to injury such as severe lower leg injury (e.g. fracture, dislocation, complete tendon rupture), requiring ridged immobilisation, or non-weight bearing status. The change in mobility should be assessed in relation to the patient’s normal state of functioning.
Venous thromboembolism	The blocking of a blood vessel by a blood clot. Includes both deep vein thrombosis and pulmonary embolism.

18. Document owner

Director Clinical Services

Enquiries relating to this guideline may be directed to:

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19. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
Version 2	02/12/2024	02/12/2027	Scheduled review

20. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Tania Strickland – A/Director Clinical Services
Authorisation date	02/12/2024
Published date	02/12/2024

Appendix 1 – Adult VTE risk assessment criteria

High Risk		Consider VTE Risk Factors	VTE Risk Factors	
☐ Total hip replacement, total knee replacement, or hip fracture surgery ☐ Abdominal or pelvic surgery for cancer ☐ Multiple major trauma ☐ Acute spinal cord injury with paresis			☐ Moderate to major * surgery *operating time > 45 minutes and/or intra-abdominal ☐ Prior history of VTE ☐ Known thrombophilia (including inherited disorders) ☐ Active malignancy or cancer treatment ☐ Significantly reduced mobility relative to normal state ☐ Age > 60 years ☐ Congestive heart failure ☐ Myocardial infarction ☐ Active or chronic lung disease ☐ Active infection ☐ Inflammatory bowel disease ☐ Active rheumatic disease ☐ Obesity (BMI > 30kg/m ²) ☐ Hormonal replacement therapy ☐ Oestrogen-based contraceptives ☐ Myeloproliferative neoplasms ☐ Pregnant or < 6 weeks post-partum ☐ Varicose veins/chronic venous stasis ☐ Nephrotic syndrome ☐ Dehydration ☐ Sickle cell disease	
Moderate Risk				
☐ Patients who are not in either the lower-or higher-risk group				
Low Risk				
☐ Ambulatory patient without VTE risk factors ☐ Ambulatory patient with VTE risk factors BUT expected length of stay ≤ 2 days ☐ Minor surgery * in patient without VTE risk factors *same day surgery or operating time < 30 mins				
Identify contraindications and other conditions to consider with pharmacological prophylaxis				
Absolute Contraindication ☐ Active haemorrhage ☐ Severe trauma to head or spinal cord, with haemorrhage in last 4 weeks ☐ Thrombocytopenia (platelets <50x10 ⁹ /L) OR coagulopathy ☐ End stage live disease (INR >1.5) ☐ Therapeutic anticoagulation with medication ☐ Other		Relative Contraindication (Consider risk vs benefit) ☐ Intracranial haemorrhage within last year ☐ Craniotomy within 3 weeks ☐ Introcular surgery within 2 weeks ☐ Gastrointestinal OR genitourinary haemorrhage within last month ☐ Active intracranial lesions / neoplasms ☐ Hypertensive emergency ☐ Post-operative bleeding concerns ☐ Antiplatelet therapy ☐ Inherited bleeding disorder ☐ High falls risk		
		Other Conditions ☐ Heparin-sensitivity or history or heparin-induced thrombocytopenia ☐ Insertion/removal of epidural catheter or spinal needle (current or planned) ☐ Creatinine clearance <30mL/min ☐ Acute stroke ☐ Neurosurgery ☐ Weight <50kg or >100kg		
Identify contraindications to mechanical prophylaxis				
☐ Skin ulceration ☐ Severe peripheral vascular disease ☐ Severe dermatitis ☐ Severe lower leg deformity		☐ Recent lower limb DVT ☐ Massive leg/pulmonary oedema due to congestive cardiac failure ☐ Morbid obesity (where correct fitting of stocking cannot be achieved)		
		☐ Peripheral neuropathy (IPC can be used) ☐ Recent skin graft ☐ Stroke patients (avoid anti-embolic stockings)		

Appendix 2 – Recommended VTE prophylaxis

Higher Risk		
Select one pharmacological option: ☐ Enoxaparin 40 mg subcutaneous daily ☐ Enoxaparin 20 mg subcutaneous daily if creatinine clearance <30mL/min (or use heparin 5000 units subcutaneous 8 or 12-hourly) ☐ Dalteparin 5000 units subcutaneous daily ☐ Unfractionated Heparin 5000 units subcutaneous three times daily ☐ Alternative oral agent for Orthopaedic surgical patients (see below)* OR ☐ No pharmacological prophylaxis due to contraindication or not advised		AND select one mechanical device: ☐ Graduated compression stockings ☐ Intermittent pneumatic compression (IPC) ☐ Foot impulse device ☐ No mechanical prophylaxis because of contraindication PLUS ☐ Early Mobilisation ☐ Patient education
Moderate Risk		
Select one pharmacological option: ☐ Enoxaparin 40 mg subcutaneous daily ☐ Enoxaparin 20 mg subcutaneous daily if creatinine clearance <30mL/min (or use heparin) ☐ Dalteparin 5000 units subcutaneous daily ☐ Heparin 5000 units subcutaneous 8- or 12-hourly OR ☐ No pharmacological prophylaxis due to contraindication or not advised		OR select one mechanical device if not prescribing pharmacological prophylaxis: ☐ Graduated compression stockings ☐ Intermittent pneumatic compression (IPC) ☐ Foot impulse device ☐ No mechanical prophylaxis because of contraindication PLUS ☐ Early mobilisation ☐ Patient education
Low Risk		
☐ Prophylaxis not required	☐ Early mobilisation	☐ Patient education
Other Considerations		
Insertion or removal of epidural catheter or spinal needle (lumbar puncture) should be carried out ≥ 2 hours before a prophylactic dose of LMWH AND ≥ 12 hours after a previously administered dose. *Orthopaedic Surgery: Alternative agents may include: ☐ <u>Hip or knee arthroplasty:</u> dabigatran, rivaroxaban, fondaparinux, apixaban or aspirin ☐ <u>Hip fracture:</u> fondaparinux or aspirin in combination with LMWH		
Duration of therapy		
Medical patients: ☐ Continue until acute medical condition is stable, patient is mobile or until hospital discharge Surgical patients: ☐ Hip fracture surgery – continue for 28 to 35 days ☐ Hip or knee arthroplasty – enoxaparin as an inpatient and aspirin as an outpatient for six weeks ☐ Lower leg immobilisation due to injury – until fully mobile ☐ Major general surgery – continue for up to 1 week or until fully mobile		

Appendix 3 – Mechanical VTE prophylaxis for surgical patients in the Same Day Unit

Please refer to the adult VTE risk assessment criteria (Appendix 1) or risk categories and contraindications.

Surgical specialty		GCS	IPC	Notes
All surgical patients	Low risk	–	–	Minor surgery in patients without VTE risk factors (same day surgery or operating time <30 mins)
Orthopaedics	Moderate risk	–	–	
	High and all lower limb joints	–	Y	Non-operative leg only for lower limb surgery
Obstetrics	All elective caesarean sections	–	Y	
General surgery including: Urology ENT Gynaecology	Moderate risk	Y	–	Use IPC if GCS are contraindicated
	High risk	–	Y	Consider GCS if IPC are contraindicated

Considerations:

- Use either Graduated Compression Stockings (GCS) **OR** Intermittent Pneumatic Compression (IPC) – **NOT** both.
- GCS or IPC garment to be fitted in the Same Day Unit.
- IPC to be connected to the machine once patient is in the operating theatre.